

Math 142, S14

Quiz 12

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) The Cartesian coordinates of a point are given.

(a) $(-1, \sqrt{3})$. Find the polar coordinates (r, θ) of the point where $r > 0$ and $0 \leq \theta < 2\pi$.

$$r = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{4} = 2$$

$$\tan \theta = \frac{y}{x} = -\sqrt{3}$$

$(-1, \sqrt{3})$ is in the 2nd quadrant $\Rightarrow \theta = \frac{2\pi}{3}$

$\Rightarrow$ polar coordinates: $(2, \frac{2\pi}{3})$

(b) $(3, -3)$. Find the polar coordinates (r, θ) of the point where $r < 0$ and $0 \leq \theta < 2\pi$.

$$r = \sqrt{3^2 + (-3)^2} = \sqrt{18} = 3\sqrt{2}$$

$$\tan \theta = \frac{-3}{3} = -1$$

$(3, -3)$ is in the 4th quadrant $\Rightarrow \theta = \frac{7}{4}\pi$

If we require $r < 0$, then its polar coordinates is $(-3\sqrt{2}, \frac{3\pi}{4})$

Problem 2. (4 points) Identify the curve by finding a Cartesian equation for the curve:

$$r = 3 \sin \theta$$

$$r^2 = 3r \sin \theta$$

$$x^2 + y^2 = 3y$$

$$x^2 + y^2 - 3y = 0$$

$$x^2 + y^2 - 3y + \left(\frac{3}{2}\right)^2 = \left(\frac{3}{2}\right)^2$$

$$x^2 + \left(y - \frac{3}{2}\right)^2 = \frac{9}{4}$$

$\Rightarrow$ a circle centered at $(0, \frac{3}{2})$ with radius $\frac{3}{2}$